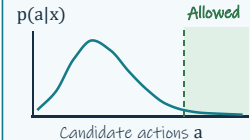


Before alignment



Reference vs. rollout

Occupancy: a separate gap



Policy mismatch

(a) Harness-Policy Teacher

Reference replay in Harness

Recorded a^* advances state; Native context x^*

AUTOPROBE $\rightarrow$ control document d^H

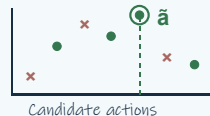
Pre-action facts + Reference ledger + Action witness

Teacher model $\pi_{\theta_0}(\cdot | \text{Inject}(x^*, d^H))$

Sample proposals



Verify & select



(b) State-conditioned Policy Alignment

HARNESS-REPLAY

$x^* \rightarrow \tilde{a}$

INJECTED

$\text{Inject}(x^*, d^H) \rightarrow \tilde{a}$

RUNTIME

$\text{Probe}(x^*) \rightarrow d^H$

Shared θ · interleaved training views

Control-entry alignment

--- Before — After

Prob Density



State-valid actions

--- Before — After

$p(a|x)$

Allowed by Harness

